Clinica Chimica Acta 548 (2023) 117519

Contents lists available at ScienceDirect

Clinica Chimica Acta

journal homepage: www.elsevier.com/locate/cca

Review

Advancing AI in healthcare: A comprehensive review of best practices

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ARTICLE INFO

Keywords:
Artificial intelligence
Machine learning
Health equity
Best practices
AI ethics
AI biases

ABSTRACT

Artificial Intelligence (AI) and Machine Learning (ML) are powerful tools shaping the healthcare sector. This review considers twelve key aspects of AI in clinical practice: 1) Ethical AI; 2) Explainable AI; 3) Health Equity and Bias in AI; 4) Sponsorship Bias; 5) Data Privacy; 6) Genomics and Privacy; 7) Insufficient Sample Size and Self-Serving Bias; 8) Bridging the Gap Between Training Datasets and Real-World Scenarios; 9) Open Source and Collaborative Development; 10) Dataset Bias and Synthetic Data; 11) Measurement Bias; 12) Reproducibility in AI Research. These categories represent both the challenges and opportunities of AI implementation in healthcare. While AI holds significant potential for improving patient care, it also presents risks and challenges, such as ensuring privacy, combating bias, and maintaining transparency and ethics. The review underscores the necessity of developing comprehensive best practices for healthcare organizations and fostering a diverse dialogue involving data scientists, clinicians, patient advocates, ethicists, economists, and policymakers. We are at the precipice of significant transformation in healthcare powered by AI. By continuing to reassess and refine our approach, we can ensure that AI is implemented responsibly and ethically, maximizing its benefit to patient care and public health.

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as vital tools in healthcare, though their implementation faces hurdles, including concerns about transparency, ethics, fairness, algorithmic biases, and privacy [1]. To address these issues, best practices in AI-based clinical applications have been developed, with significant contributions from The Lancet Commission and the World Health Organization (WHO) [2,3] (Table 1). In particular, the WHO identified six core principles for AI in healthcare: (1) Protect autonomy; (2) Promote human well-being, human safety, and the public interest; (3) Ensure transparency, explainability, and intelligibility; (4) Foster responsibility and accountability; (5) Ensure inclusiveness and equity; (6) Promote AI that is responsive and sustainable [3].

Various organizations have striven to establish best practices in AI healthcare as summarized in Table 1 and include recommendations for initiatives, frameworks, guidelines and education.

This article reviews recent developments in best practices for AI in clinical practice and provides critical commentary.

The FDA, the EU and regulators around the world are struggling to determine how to better address recent AI advancements. Large language models (LLM) such as OpenAI's ChatGPT and Google's Med PaLM have shown potential in healthcare applications. Yet, their utilization is hindered by ethical considerations. These models lack a source of 'ground truth', making the validation of their performance challenging, and rigorous and systematic testing is suggested to enhance these models' focus and accuracy [18]. In the context of data science, machine learning, and artificial intelligence, 'ground truth' refers to the absolute, definitive reality or the factual information that is used as a reference to train and test a model.

Regulatory reaction has mostly been reactive, as it tries to catch up with rapidly advancing progress in artificial intelligence, so much so, that the current practices in digital health have been described as the "Wild West". [19].

The EU's draft AI Act contains obligations concerning the accuracy, robustness and cybersecurity of systems themselves, with particular regard to discrimination, as systems learn, as well as adversarial machine learning. However, there is no explicit discussion of leakage of training data or other personal data from models.

Similarly, instead of focusing on explainability, the FDA and other regulators should closely scrutinize the aspects of AI / ML that affect patients, such as safety and effectiveness, and consider subjecting more health-related products based on artificial intelligence and machine learning to clinical trials [20].

As regulators strive to keep pace with the advancements in AI, we should avoid allowing AI models to self-regulate [21]. It is imperative that AI companies demonstrate best practices in the development of their models.

Academic studies offer some guidance on how to implement ML health projects ethically and responsibly. For example, a 'pipeline framework' to facilitate interdisciplinary dialogue and manage ethical implications of ML in healthcare has already been developed in

https://doi.org/10.1016/j.cca.2023.117519
Received 2 July 2023; Received in revised form 14 August 2023; Accepted 15 August 2023
Available online 16 August 2023
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Table 1
Best practices in AI healthcare initiatives, in chronological order.

Date	Organization	Summary	References
2016	U.S. Federal Trade Commission	Report highlighted the importance of avoiding algorithmic bias.	[4]
2017	Stanford Human-Centered AI institute	Annual AI Index report monitors AI developments and their ethical implications.	[5]
2019	Brookings Institution	'AI Governance' educational series addresses challenges associated with emerging technologies.	[6]
2019	European Commission High-Level Expert Group on AI	Outlined seven key requirements that AI systems should meet as part of its Ethics Guidelines for Trustworthy AI.	[7]
2019	Organisation for Economic Co-operation and Development (OECD)	Adopted its Principles on Artificial Intelligence, the first international standards agreed by governments for the responsible stewardship of trustworthy AI.	[8]
2021	European Commission	Proposed AI Act with the idea that the use of artificial intelligence can support socially and environmentally beneficial outcomes and provide key competitive advantages to the European companies and the economy. AI technologies would be classified into 4 different categories of risk. High-risk AI systems in areas such as healthcare would be strictly regulated.	[9]
2021	World Health Organization (WHO)	Guidance report on Ethics & Governance of Artificial Intelligence for Health identifies six core principles to ensure AI works to the public benefit of all countries.	[3]
2021	U.S. Food and Drug Administration (FDA), Health Canada, and the UK's MHRA	Identified 10 guiding principles that can inform the development of Good Machine Learning Practice (GMLP).	[10]
2021	The Lancet Commission	Report on the future of digital technologies in healthcare, including racial discrimination and ethical challenges caused by such technologies.	[2]
2021	UK Health Data Research Alliance	Published "Principles and Best Practices for Trusted Research Environments" that set out an approach for a trustworthy ecosystem of data access for health research.	[11]
2022	The Artificial Intelligence Industry Innovation Coalition (AI3C)	AI3C set up to serve as a source of best practices for AI in healthcare, including responsible AI implementation, health equity, and transformation to reduce clinical fatigue.	[12]
2023	National Institute of Standards and Technology (NIST) of the U.S. Department of Commerce	Launched the Artificial Intelligence Risk Management Framework (AI RMF 1.0) to offer a resource to the organizations designing, developing, deploying, or using AI systems to help manage the many risks of AI and promote trustworthy and responsible development and use of AI systems.	[13]
2023	U.S. Food and Drug Administration (FDA)	Draft guidance outlines a new process in which the developers of AI tools could get approval for modifications in advance by submitting a document describing how the changes will be implemented and tested.	[14]
2023	World Association of Medical Editors (WAME)	Recommendations for the use of ChatGPT and other chatbots in scholarly publications, including the need for journal editors to adopt new tools to detect AI-generated or modified content.	[15]
2023	United Nations (U.N.)	U.N. Secretary-General backed a proposal for the creation of a high-level AI advisory body to regularly review AI governance arrangements and offer recommendations on how they can align with human rights, the rule of law and common good.	[16]
2023	American Medical Association (AMA)	Agreed to develop principles and recommendations concerning the benefits and potential unforeseen consequences of relying on AI-generated medical advice and content that may not be validated, accurate, or appropriate.	[17]

academia [22], and recommendations proposed that span the pipeline stages of a typical ML health project [23]. We must also encourage the establishment of AI ethics committees to manage the ethical risks of AI products, ensuring diversity in committee composition [24].

The need for a regulatory body in AI is undeniable. However, we must not overlook crucial contributors such as AI ethics organizations, patient advocacy groups and social workers, whose insights are vital for addressing human-centered ethical challenges. Prioritizing technology over human input from these groups risks repeating past mistakes and impedes true innovation.

2. AI in clinical practice

Twelve key aspects of AI in clinical practice have been chosen for review. These aspects include:1. Ethical AI in Clinical Practice; 2. Explainable AI, or Interpretable AI; 3. Health Equity, Racial/Gender, and Algorithmic Biases in Artificial Intelligence; 4. Sponsorship Bias in Medical Literature; 5. Data Privacy in Clinical Practice; 6. Genomics and Privacy Protection; 7. Insufficient Sample Size, Inappropriate Level of Statistical Significance (Alpha), and Self-Serving Bias; 8. Bridging the Gap Between Training Datasets and Real-World Scenarios; 9. Fostering a Culture of Open Source and Collaborative Development in AI Applications in Healthcare; 10. Dataset Bias, Synthetic Data, and Data-Centric AI; 11. Measurement Bias; and 12. the Reproducibility or Replicability Crisis in AI Research.

2.1. Ethical AI in clinical practice

Integrating AI into healthcare is complex yet potentially transformative. However, as we navigate this landscape, we must confront ethical dilemmas, placing greater emphasis on informed consent, patient privacy, and the role of physicians in AI-mediated care, as stressed by the AMA Journal of Ethics [25]. While several legal and ethical challenges have been highlighted [26], the influence of AI on healthcare intellectual property law warrants more in-depth exploration. This aspect could critically shape the future of AI in healthcare. While the developed ethical frameworks for Large Language Models (LLM) in healthcare AI are comprehensive [27], there is a need for broader consideration of societal implications. The most recent analysis of AI in clinical medicine is thorough [28], but it may not sufficiently address potential misuse and AI bias. Future research should prioritize the establishment of measures against these risks. I wholeheartedly join other academics in advocating for collaboration between AI developers, healthcare professionals, and ethicists [29]. This alliance is crucial for responsible AI development and implementation, a sentiment that has been reinforced by others [30], especially in relation to LLM.

Achieving ethical AI in clinical practice requires careful planning and adherence to several critical principles. The essential steps to ensure the ethical use of AI in healthcare encompass: formulating and complying with ethical guidelines, maintaining transparency and explainability, honoring patient privacy and confidentiality, and engaging key stakeholders such as patients, healthcare providers, and ethicists in the development process. I will elaborate on some of these steps in subsequent sections.

Numerous ethical guidelines have been put forth for the implementation of AI in clinical practice. For instance, the World Health Organization (WHO) has released its inaugural global report on AI in health, outlining six guiding principles for its design and utilization [3]. This report emphasizes that AI has the potential to revolutionize healthcare and medical services globally, but its success depends on focusing on ethics and human rights in its design, implementation, and usage.

Similarly, the American Medical Association (AMA) has crafted a framework for the creation and application of AI in healthcare. This approach builds on the AMA policy for augmented intelligence, informed by the most recent research and filtered through the considerations of ethics, evidence, and equity. The framework is termed Trustworthy Augmented Intelligence in Health Care [118].

Furthermore, the ongoing conversation about ethical AI in healthcare is heading in a positive direction, but there needs to be an intensified focus on issues such as intellectual property rights and bias mitigation. This refined attention will further strengthen the ethical foundation of AI in clinical practice.

2.2. Explainable AI, or interpretable AI

AI, often viewed as a "black box", faces an "explainability crisis", particularly in clinical settings where understanding AI decisions is crucial [31]. While some argue for transparency into every algorithmic detail, others, echo Cassie Kozyrkov at Google, and suggest a need for general rather than comprehensive understanding, analogous to using a microwave oven [32]. Despite data scientists' differing views, legislation must guard against potential discriminatory consequences arising from combining AI with health data [33].

The "Explainable AI in Healthcare and Medicine" book underlines the importance of explainable AI models for transparency and accountability in healthcare [34]. However, interpretability is not straightforward. Feature attribution methods attempt to measure the significance of individual inputs to a model, but such post-hoc explanations can overlook crucial factors or misinterpret the model's focus [35].

A different crisis is that the current explanations provided by AI systems are often inadequate or misleading [36]. Some researchers suggest that stringent explainability requirements might stifle innovation, urging regulators to focus on safety and effectiveness [20]. A shift towards interpretable models has been proposed, based on the argument that they perform comparably to black box models, provided they are optimally utilized [37]. However, there is a danger of spuriously assigning causality in post-hoc interpretations, and the pitfalls of relying on two models - the black box and the explanation model [38]. This distinction between post-hoc explanations and standalone interpretable models is vital in fields like healthcare where stakes are high.

Thus, while AI continues to advance, its integration into clinical practice necessitates ongoing efforts to enhance its explainability and interpretability. A concerted, collaborative approach between AI developers and clinicians should bridge this gap, ensuring AI tools are user-friendly, trustworthy, and fit for purpose in the healthcare context [39].

Ultimately, the imperative for interpretable AI models is greater in healthcare than other sectors due to the life-altering consequences of errors. The challenge remains for end-users to distinguish between the post-hoc explained models and standalone interpretable models.

2.3. Health equity, racial / gender and algorithmic biases in artificial intelligence

Racial and gender biases in medical practice are well-documented and often intensified by AI [40–45]. The 2016 Arkansas Medicaid algorithm serves as a chilling reminder of how these biases can create systemic injustices, failing to properly account for needs of patients with cerebral palsy [46]. A UnitedHealth Group algorithm investigated in 2019 exemplifies how race can influence AI decision-making, even when not an explicit factor. Economic disparities reflected in health costs unintentionally led to the prioritization of healthier white patients over sicker black patients [47,33].

I advocate the concept of 'algorithmovigilance', inspired by the so-called "pharmacovigilance" [48]. It underscores the need for continual evaluation of algorithms to ensure fairness and efficacy. Eliminating biases requires vigilance at every stage of an AI system's development. These biases, often unconsciously introduced by humans, include historical bias, representation bias, measurement bias, and coding bias [47].

Algorithms need diverse, comprehensive data sets to mitigate regional disparities in healthcare [49]. However, AI's potential harm extends beyond healthcare. For instance, flawed language models can propagate dangerous misinformation [50,51]. Our responsibility is to strive for AI fairness, diligently working to prevent racial and gender disparities, such as those found in the Obermeyer et al. study [33].

Racial, gender, and algorithmic biases in Artificial Intelligence (AI) exacerbate healthcare inequities. Initiatives to de-bias AI, including IBM's AI Fairness 360 toolkit [52] and MIT's Partial Attribute Decorrelation [53], aim to mitigate these disparities. However, a more proactive approach considers ethical and governance issues before AI model development [54]. Furthermore, Stanford researchers argue for modifying data collection policies to tackle bias [55,56]. In this vein, 'value sensitive design' calls for early stakeholder involvement and careful dataset curation [57]. It also suggests recognizing the role of AI in health care, particularly in primary care [58].

Nevertheless, AI must cater to the diversity of individuals, including those who identify beyond the binary gender spectrum. Current digital health systems (DHS) are inadequate in capturing nuanced gender, sex, and sexual orientation (GSSO) data [59]. Thus, DHS must be designed inclusively to ensure health equity.

Regulatory strategies, involving bias mitigation throughout the AI lifecycle, are also necessary to uproot bias in healthcare AI [60]. Large societal problems underlying systemic inequalities need to be addressed to realize the potential of AI in health equity [61]. Ultimately, addressing bias in healthcare AI, must prioritize serving the common good over profit [62]. This approach is crucial to avoid coded biases – those biases that are inadvertently integrated into AI models – and to ensure equitable outcomes in healthcare.

While AI holds promise for reducing health disparities, efforts towards social justice and structural changes remain vital. One example is fostering AI applications in primary care, focusing on infrastructure updates, delivery transformation, evaluation modernization, algorithm marketing authorization, reimbursement, and social justice and health equity [58].

2.4. Sponsorship bias in medical literature

Sponsorship bias, a long-standing phenomenon in medical literature, raises profound concerns about the integrity of clinical studies. Typically, sponsored studies demonstrate a tendency to produce results favoring the sponsor [63]. Notably, this bias is manifest in both in manuscripts and peer-reviewed studies, suggesting a ubiquitous issue within the scientific community. Studies suggest that industrial sponsorship of drug and device studies often yields more favorable results than those funded by other sources [63]. Meanwhile, medical communication companies, including 'symptom checkers' like WebMD and Medscape, frequently receive substantial funding from pharmaceutical and device companies, further entrenching the bias [64].

Clinical research, given its profound implications for human lives, should uphold stringent standards of scientific integrity. However, economic incentives can engender conflicts of interest, potentially leading to the withholding of unfavorable results or even data tampering [65]. Large pharmaceutical companies may possibly exert undue influence over academic research and clinical trials, potentially skewing results and suppressing innovation [66].

Alleviating commercial bias requires concerted effort. One approach involves establishing a robust firewall between funding and research conduct, while another proposes creating an independent funding source. Yet, implementing these measures remains a practical challenge [67]. It is essential that we collectively strive for transparency and unbiased reporting in clinical research, thereby safeguarding scientific credibility and patient well-being.

2.5. Data privacy in clinical practice

Data privacy remains paramount in healthcare, especially given the sensitivity of personal health information. Researchers involved in human studies must ensure robust data security measures, including stringent server backtesting and third-party provider diligence [68].

Ethical and human rights implications surrounding the collection of health data, has sparked debates over the infringement of privacy for broader purposes such as digital contact tracing for COVID-19 [68]. Similar concerns arise with AI technologies like facial recognition, considered by some to be invasive and potentially biased [69]. To address these, Microsoft recently committed to enhancing control over its AI products, including withdrawing certain facial analysis features [70].

Simultaneously, the rising popularity of AI chatbots, such as ChatGPT, and digital health apps engenders concern over data usage and sharing. Users might disclose sensitive health information to AI, unaware that it may be exploited for business purposes, even if stated in privacy agreements [71]. The current state of digital health apps walks a thin line between empowerment and manipulation [1,72]. Furthermore, the dilemma between individual and collective interests intensifies as data-driven healthcare promises increased public welfare benefits while potentially infringing upon privacy rights [73]. This tension underscores the need for open debates and regulation to ensure a balanced approach to healthcare data privacy.

2.6. Genomics and privacy protection

Genomic research, while promising remarkable advancements in health and treatment, poses significant challenges to privacy and informed consent models. A critical debate initiated in 2001 questioned the duty of the public to contribute data to genetic databases [74]. This debate remains relevant today, as "biobanks" have evolved, enabling data linkage and mining beyond the constructs of traditional research ethics. Furthermore, the comprehensive ethical and human rights implications of evolving genomic technologies remain uncharted territory. It is indisputable that all forms of privacy warrant protection; however, the genome demands extraordinary respect. The viewing of a genome has been equated to witnessing nudity, a comparison underscoring the deep intimacy and privacy associated with genomic data [75].

The legal consequences of deidentifying shared genomes have also been studied, as have how these deidentified genomes could potentially be reidentified, and their information used detrimentally [76]. Therefore, it is paramount that future discussions and policies incorporate these considerations to foster public trust in genomic research.

2.7. Insufficient sample size, inappropriate level of statistical significance (alpha), and self-serving bias

It is disconcerting to witness the frequent neglect of adequate sample sizes and inappropriate levels of statistical significance in many published clinical studies, particularly concerning medical devices. In key peer-reviewed studies on digital symptom checkers, for instance, over half rely on fewer than 100 vignettes for their analyses, with one employing a mere 28 [77–82]. Given current computational capabilities that allow for the simulation of thousands of vignettes across a myriad of diseases, it is baffling that studies continue to draw from such limited samples, seemingly overstating the statistical significance of these constrained subsets.

In terms of best practices related to sample size and statistical significance, the American Statistical Association (ASA) emphasizes the importance of reporting effect size and the confidence interval of the estimated statistics to properly understand the practical relevance or clinical significance [119]. Moreover, in the era of big data, considerations around the impact on the p-value must be acknowledged, and the estimation and reporting of effect size should be aligned to add value in this context. This approach provides a more comprehensive understanding of the results, and the potential magnitude of the findings, rather than focusing solely on whether a result is statistically significant [120].

Parallel to this issue, a prevalent self-serving bias in empirical research can be discerned, akin to how media companies frequently highlight crime and crises to cater to public interest, thereby warping perceptions of reality. Another example of self-serving bias is patients' doctor-shopping until they receive a diagnosis that matches their expectations [83].

A more specific form of self-serving bias known as p-hacking must also be highlighted. This includes practices such as "data shopping" — seeking out data that corroborates preconceived notions while discounting the rest. It also encompasses other types, such as conducting multiple analyses and reporting only those that yield significant results, and stopping data collection once significance is reached [121]. Such practices obstruct the path to scientific truth and are a form of self-serving quest for validation that has become alarmingly common.

I propose the following recommendations to overcome the above-mentioned issues of insufficient sample size, inappropriate level of statistical significance (alpha), self-serving bias, and p-hacking.

- Incorporating effect size and confidence intervals: Based on the ASA's recommendations, researchers should consider the effect size and confidence intervals to provide context to their statistical results. This includes understanding the practical relevance or clinical significance of findings, not just whether they are statistically significant [119].
- Utilizing appropriate sample sizes: Leveraging computational capabilities to simulate realistic and robust sample sizes will create a more valid and reliable understanding of medical phenomena. This ensures that the results are applicable and representative of the population of interest [122].
- Avoiding p-hacking and other self-serving biases: Transparency in data analysis and reporting, pre-registering research plans, and adopting robust research ethics will minimize the risk of p-hacking and other self-serving biases. By promoting integrity in research processes, the scientific community can work towards a more authentic and reliable body of knowledge [123].

By actively embracing these recommendations, the medical research community can advance towards more valid, ethical, and unbiased research. These principles are not only theoretical but translate directly into practical strategies that can be implemented in real-world clinical settings, thereby enhancing the quality and credibility of scientific inquiry in the field of AI in healthcare.

2.8. Bridging the gap between training datasets and real-world scenarios

AI's potential for reducing diagnostic errors remains marred by inherent discrepancies between training data and real-world clinical scenarios [84]. This gap is due to limited high-quality training data and AI's inability to identify contextual changes that may affect the validity of learned predictive assumptions. Consequently, AI might apply predetermined methodologies inappropriately, leading to errors. Also, unlike human clinicians, AI lacks the training to err on the side of caution, potentially impacting patient safety.

Research by Stanford University revealed significant weaknesses in medical AI devices approved by the FDA. In a sample of 130 devices, 126 were evaluated using retrospective data, devoid of active human clinician input. Furthermore, demographic data reporting, such as sex, gender, or race/ethnicity, was conspicuously absent in over 87% of approved device performance summaries [85,56].

A pertinent issue related to these discrepancies is the importance of choosing appropriate metrics when dealing with imbalanced data. Standard metrics, such as accuracy, may be misleading in medical

contexts where data may be imbalanced. Utilizing metrics such as precision, recall, or the F1 score can offer a more nuanced understanding of a model's performance, particularly for minority classes [124].

The lack of generalizability, or overfitting, presents another challenge. AI models tend to perform well statistically but fail in real-world clinical settings. Overfitting can be mitigated by employing techniques like regularization, cross-validation, and early stopping. These approaches can enhance a model's ability to generalize by preventing it from fitting too closely to the training data [125,126]. Alarmingly, one early warning system for sepsis was found to miss two-thirds of cases, issue false alarms, and rarely detect cases overlooked by medical staff [86]. These models also exhibited geographic inconsistencies and decreasing accuracy over time, the latter termed "calibration drift" [87].

Active learning and federated learning present promising avenues to improve the generalization of AI models to real-world scenarios. Active learning involves models iteratively selecting the most informative examples to be labeled, which can make the training process more efficient [127]. Federated learning allows models to be trained across multiple decentralized devices holding local data, fostering more diverse and representative training datasets [128].

These issues may be addressed by promoting the concept of algorithmic hygiene [88]. This practice involves comprehending bias sources and ensuring maximum representation in training data. However, no dataset can represent all possible scenarios, highlighting the need for clear target applications and audience identification before tailoring the training data.

Addressing the identified challenges requires a multifaceted approach that considers the nuanced selection of metrics, implements specific strategies to prevent overfitting, and leverages emerging techniques like active learning and federated learning. These measures can bridge the gap between training datasets and real-world clinical scenarios, thereby enhancing AI's effectiveness in healthcare.

2.9. Fostering a culture of open source and collaborative development in AI applications in healthcare

The culture of collaboration and code sharing among data scientists and developers, uncommon in other industries, has significantly expedited AI research and innovations. Standards enabling transparency, fairness, and data sharing in AI applications to healthcare are essential while concurrently safeguarding patient privacy [89]. Practices such as participant-centered development of AI algorithms, responsible data sharing, and code sharing, which can synthesize under-represented data to address bias are to be encouraged.

The application of these principles was effectively demonstrated in the CORD-19 project against COVID-19, organized by the White House and a consortium of research groups, and managed by Kaggle [68,90]. These open-source research competitions enhance consumer trust and pinpoint the most appropriate technologies for specific health scenarios. However, the competitive element can occasionally eclipse the collaborative aspect. Certain individuals and organizations might hesitate to share their best ideas, apprehensive about providing their competitors with an advantage.

Stanford University researchers posit that AI and sensor-based "ambient intelligence" can enhance outcomes and avoid medical errors. They stress the need for multi-disciplinary collaboration to develop trustworthy ambient intelligence systems in healthcare [91].

In imaging and computer vision, the advancement of image recognition technology has surpassed human visual capability. The origin of these innovative technologies is frequently found in open-source codes, often provided by major tech firms. AI can streamline the diagnosis and treatment decision-making process by integrating imaging results into electronic medical records. It could even predict the likelihood of success of different treatment protocols [92]. The significance of this approach is underscored by the fact that 20,000 to 40,000 deaths occur annually due to misdiagnosis, with over 70% of treatment plans suboptimal. No human can manage the predicted doubling of medical knowledge every 73 days. However, computers and NLP systems can, thus aiding in preventing misdiagnosis and death [93].

It is important that medical organizations, corporations, and commercial laboratories to be open to independent testing and to participate in research competitions for the greater benefit of clinicians and patients.

2.10. Dataset bias, synthetic data and data-centric AI

Throughout the evolution of AI, emphasis has primarily been on enhancing coding and modeling. However, the datasets used to develop and train these models often remain raw, unstructured, and unchecked. This laxity in understanding the data leads to persistent biases and ethical dilemmas [94].

Bias manifests within healthcare datasets when there is systematic disregard or denial of care based on a patient's race or when data reflects only those who can afford care [94]. It has been demonstrated that face recognition algorithms are less efficient for "darker females" due to underrepresentation in training data [69]. Consequently, there is a data diversity issue in medical AI: researchers struggle to acquire large, diverse medical datasets, culminating in biased algorithms [95].

Another problem associated with dataset bias is distributional disparities in patient and physician populations within US healthcare datasets. Electronic Health Record (EHR) / Electronic Medical Record (EMR) systems, from which these datasets are extracted, often reflect this bias [96]. Moreover, human errors in data entry contribute to the so-called "dirty" datasets, which are raw, unformatted, and often fraught with missing fields. Such biases have severe implications for AI-based diagnostic or prognostic methods. Statisticians refer to this as the Garbage In Garbage Out (GIGO) concept. Addressing these biases could be achieved via the "de-biasing" algorithms proposed by IBM and Microsoft [52] or through a careful and selective approach to data [97].

Despite the prevalence of pre-trained models, it is crucial to understand the quality and accuracy of the underlying data. This understanding underscores the significance of synthetic data, especially in scenarios where training data is scarce [98]. Synthetic data can increase dataset diversity and enhance the robustness and adaptability of AI models. However, it is important to consider potential malicious uses of synthetic data, such as deepfakes [99].

Recently, the concept of data-centric AI was introduced, emphasizing the systematic engineering of data for successful AI development [100]. While not entirely novel, this approach gives us the ability to engineer a subset of the data, thus addressing biases in a more targeted way [101,102].

Addressing dataset bias is a step towards improving AI models, which is vital in healthcare where the health and lives of patients are at stake.

2.11. Measurement bias

Measurement bias in medical research can significantly undermine the integrity of results, even in the presence of rigorous algorithmic and dataset bias mitigation. Clearly, without appropriate consideration for this bias, conclusions may be misleading or erroneous as evidenced by a study by researchers from Stanford University in 2017. This evaluated the efficacy of wearable heart rate monitors, such as Fitbit, that have been used in more than 300 clinical trials [103]. These devices predominantly utilize green light technology to measure heart rate, offering a cost-effective means for data collection. It was found, however, that this technology demonstrated less accuracy in individuals with higher levels of skin melanin, indicating a potential measurement bias against people of color [103]. This example underscores that, while ensuring diverse participant inclusion in clinical trials is crucial, it does not alone rectify research issues if the measurement methods applied are systematically biased. Consequently, medical research must incorporate robust strategies for the identification and correction of potential

measurement bias, further enhancing the validity and generalizability of findings.

2.12. Reproducibility or replicability crisis in AI research

Reproducibility, the capacity for independent replication of a study's results, is a cornerstone of empirical research integrity, particularly in AI studies [104]. Regrettably, many published studies face a reproducibility crisis, initially noted in industry research and now permeating academia [105,106].

Reproducibility in empirical AI research is predicated on the ability of an independent research team to replicate results solely based on the original team's documentation [104]. Several authors have raised concerns related to credibility, reproducibility, and documentation in AI-based research, stressing the necessity for reproducible research outcomes [107–111].

Despite academic research reportedly outperforming industry on several reproducibility metrics, the general consensus highlights the need for improving empirical AI research documentation [108]. As a countermeasure, an author checklist for documenting AI research, guided by best practices from various scientific entities, has been introduced [109].

The checklist [109] specifically addresses the factors required to facilitate reproducibility in AI research, such as an explicit mention of the problem being solved, research objectives, research methods (whether empirical or theoretical), research questions, AI method pseudocode, sharing of training, validation, and test sets, sharing relevant results, explicit mention of hypotheses and predictions, availability of open-source code, specification of hardware and software dependencies, and sharing of variable settings like hyperparameters.

By adhering to these yes/no variables, researchers can ensure a structured framework for reproducibility, facilitating independent verification of their findings. This aligns with the broader scientific community's efforts to establish standard protocols and share critical research components [109].

It was further proposed to bolster health data sharing and reproducibility by using synthetic data that closely mimics real-world observational data [112]. However, concerns persist regarding quality and reproducibility [113]. The importance of stringent scrutiny in healthcare quality improvement studies and the prevention of bias in research conclusions is emphasized, highlighting the crucial role of reproducibility in the advancement of medical research.

Addressing these challenges involves embracing transparency, standardizing the reporting process, and providing access to key research artifacts, such as datasets, codes, and methodologies. The concerted effort towards better documentation of the research process and adherence to best practices will help mitigate the reproducibility crisis and strengthen the credibility of AI research in healthcare.

3. Conclusion

As with any innovation, the paramount goal of AI should center on human well-being, advancing patient care and community health. A vigilant approach must scrutinize AI applications in healthcare for underlying corporate, political, or inequitable agendas, even those ostensibly beneficial to patients.

Achieving ethical AI in clinical practice has been guided by frameworks such as the World Health Organization's six principles for AI design and utilization, emphasizing ethics and human rights, and the American Medical Association's Trustworthy Augmented Intelligence in Health Care framework. These guidelines illustrate the global commitment to ethics, evidence, and equity in healthcare AI. This adherence to ethical principles is key to the success of AI in revolutionizing healthcare, stressing the importance of ethical guidelines, transparency, privacy, stakeholder engagement, and focus on areas like intellectual property rights and bias mitigation.

AI can revolutionize healthcare — ethical, fair, open, unbiased, transparent, and responsible AI will retain its humanistic core, providing empathy to patients, nurses, and physicians. Yet, it must be remembered that humans require human empathy during times of discomfort and pain. Technology, including AI and machine learning, can assist physicians handle critical situations more accurately and efficiently. We, as a medical research community, must ensure physicians' comfort and confidence in using these tools by making them ethical, transparent, and explainable.

Despite healthcare's traditional resistance to change, AI can contribute significantly if it embodies best practices of ethics and bias resolution. In this context, there should be continuing advocacy for best practices, and it is also imperative to contemplate potential policy directions. I propose the following questions to guide future research and policy development: (1) How can AI better combat urgent global health crises and health injustices? (2) How can we motivate AI developers to integrate ethical considerations from the start? (3) How can we translate audits of AI systems for bias and clinical impact into meaningful clinical practice? (4) When and how should sensitive attributes, such as race and gender, be used in the analysis? (5) How can AI empower all patients ethically, safely, and equitably? [2,3,23].

"Simpler" and "smarter" interpretable models have been successful in various medical domains, demonstrating superior transparency and interpretability compared to complex deep learning models [37,114], and both the importance and aesthetic appeal of interpretable models have been emphasized [37].

It appears evident that AI will not replace human clinicians but will augment their care-giving capabilities. As AI is merely another tool, clinicians may increasingly rely on human skills such as empathy and holistic integration. The only clinicians who may face obsolescence are those resistant to AI [92].

In preparing to utilize AI in healthcare, leaders must ensure ethical data access, leverage domain expertise for dataset analysis, provide ample computing power for real-time decision making, and conduct thorough research into practical implementation to embed trustworthy AI algorithms into effective workflows. [115].

If AI is viewed as critical to healthcare's future, then a diverse group of stakeholders must be engaged in the dialogue, not just data scientists and engineers, but also clinicians, patient advocates, ethicists, economists, and policymakers, and the necessity for domain expertise and wider discussions around ethical AI has been highlighted [116,1].

Given the nascent stage of artificial intelligence (AI) integration in healthcare, we are in the initial phases of establishing comprehensive best practices for organizations to adhere to when incorporating this technology. These practices will, undoubtedly, encompass the forging of strategic partnerships and comprehending the limitations inherent to AI [117]. It is essential that we continually reassess and refine these best practices to ensure the responsible, ethical, and effective application of AI in healthcare.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

Sergei Polevikov (SP) thanks Larry Kricka (LK) for editorial reviews and comments. SP and LK are members of the International Federation of Clinical Chemistry (IFCC) Emerging Technology Division Working Group on Artificial Intelligence in Genomic Diagnosis (WG-AIGD).

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